Procurement Spend Analysis

1. Project Overview

This project focuses on analyzing organizational procurement data to understand spending patterns, supplier dependency, pricing inefficiencies, and cost optimization opportunities. An end-to-end analytics pipeline was implemented using Python for data preparation and exploratory analysis, PostgreSQL for structured querying, and Power BI for interactive visualization and business reporting.

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2. Dataset Overview

• **Total Records:** 500
• **Total Columns:** 9

Key Attributes

• **Transaction Details:** Transaction ID, Item Name, Category
• **Cost Metrics:** Quantity, Unit Price, Total Cost
• **Procurement Context:** Supplier, Buyer, Purchase Date

	TransactionID	ItemName	Category	Quantity	UnitPrice	TotalCost	PurchaseDate	Supplier	Buyer
0	TXN001	Desk Chair	Furniture	10	113.15	1131.50	2024-04-19	TechMart Inc.	Kelly Joseph
1	TXN002	Stapler	Office Supplies	16	12.62	201.92	2024-07-06	CloudSoft Corp.	Kelly Joseph
2	TXN003	Annual Software License	Software	1	5649.34	5649.34	2024-09-10	TechMart Inc.	Kelly Joseph
3	TXN004	Notepad	Stationery	13	2.92	37.96	2024-01-21	FurniWorks Ltd.	Luis Holland
4	TXN005	Notepad	Stationery	19	1.39	26.41	2024-02-03	TechMart Inc.	Cynthia Jenkins

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3. Data Quality Assessment

• The dataset was complete with **no missing or duplicate records**
• Data types were verified and corrected where required

• Purchase date was standardized to datetime format
• Numerical columns were validated for analytical consistency

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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   TransactionID  500 non-null    object 
 1   ItemName       500 non-null    object 
 2   Category       500 non-null    object 
 3   Quantity       500 non-null    int64  
 4   UnitPrice      500 non-null    float64
 5   TotalCost      500 non-null    float64
 6   PurchaseDate   500 non-null    object 
 7   Supplier       500 non-null    object 
 8   Buyer          500 non-null    object 
dtypes: float64(2), int64(1), object(6)
memory usage: 35.3+ KB
```

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4. Data Preparation & Exploratory Data Analysis (Python)

4.1 Python-Based Data Processing

The dataset was imported and explored using Pandas to understand its structure and distribution.

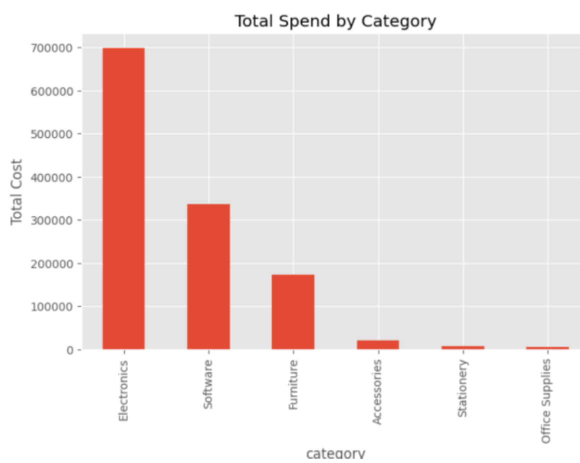
• Used df.info() and df.describe() for initial inspection
• Converted purchase dates into datetime format
• Ensured numerical fields were correctly typed
• Verified derived fields such as total cost for accuracy

	Quantity	UnitPrice	TotalCost
count	500.000000	500.000000	500.00000
mean	9.640000	854.245100	2481.16106
std	6.168834	2132.890832	3527.35316
min	1.000000	1.060000	1.56000
25%	4.000000	14.380000	145.92750
50%	10.000000	143.595000	918.56000
75%	15.000000	287.925000	3123.56750
max	20.000000	9909.240000	18494.60000

4.2 Exploratory Insights from Python EDA

• Transaction costs are **right-skewed**, indicating a small number of high-value purchases
• Strategic categories such as software and furniture dominate overall spend
• Buyer-level spending behavior varies significantly

These findings guided deeper SQL-based analysis.



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5. Database Design & SQL Analysis

The cleaned dataset was loaded into **PostgreSQL** to enable scalable querying and business-focused analysis. SQL was used to answer key procurement questions aligned with stakeholder needs.

5.1 Total Procurement Exposure

Business Question:
What is the total procurement exposure and how is it distributed?

Analysis Performed:
• Calculated total organizational spend
• Analyzed spend contribution by category
• Measured percentage share of each category

Insight:
Procurement spend is heavily concentrated in a few categories, indicating potential areas for targeted cost control.

- **Total Spend**

SELECT SUM(total_cost) AS total_spend

FROM spend_data;

	total_spend double precision
1	1240580.53

- **Spend by category**

SELECT category,

SUM(total_cost) AS category_spend,

ROUND(

(

SUM(total_cost) * 100

/ SUM(SUM(total_cost)) OVER ()

)::numeric,2

) AS spend_pct

FROM spend_data

GROUP BY category

ORDER BY category_spend DESC;

	category text	category_spend double precision	spend_pct numeric
1	Electronics	697805.2300000002	56.25
2	Software	336017.9999999999	27.09
3	Furniture	172818.61	13.93
4	Accessories	20857.24	1.68
5	Stationery	6753.1799999999985	0.54
6	Office Suppli...	6328.270000000004	0.51

5.2 Supplier Spend & Dependency Analysis

Business Question:
Which suppliers are driving procurement spend?

Analysis Performed:
• Ranked suppliers by total spend
• Calculated supplier dependency percentage

Insight:
A small number of suppliers account for a large share of total spend, increasing supplier dependency risk.

- **Supplier Spend Ranking**

```
SELECT
  supplier,
  SUM(total_cost) AS supplier_spend
FROM spend_data
GROUP BY supplier
ORDER BY supplier_spend DESC;
```

	supplier text	supplier_spend double precision
1	TechMart Inc.	328761.7300000001
2	QuickDeliver Ltd.	285353.58999999997
3	OfficeSupplies ...	223580.85
4	FurniWorks Ltd.	202810.72000000003
5	CloudSoft Corp.	200073.64000000007

- **Supplier Spend Share (% Dependency)**

```
SELECT
  supplier,
```

```
  SUM(total_cost) AS supplier_spend,
  ROUND((
    SUM(total_cost) * 100.0 / SUM(SUM(total_cost)) OVER ())::numeric,  2
  ) AS spend_share_pct
FROM spend_data
GROUP BY supplier
ORDER BY spend_share_pct DESC;
```

	supplier text	supplier_spend double precision	spend_share_pct numeric
1	TechMart Inc.	328761.7300000001	26.50
2	QuickDeliver Ltd.	285353.58999999997	23.00
3	OfficeSupplies ...	223580.85	18.02
4	FurniWorks Ltd.	202810.72000000003	16.35
5	CloudSoft Corp.	200073.64000000007	16.13

5.3 High-Risk / High-Value Transactions

Business Question:
Which transactions represent high financial risk?

Analysis Performed:
• Identified transactions exceeding defined cost thresholds
• Calculated average transaction value

Insight:
High-value transactions are infrequent but significantly impact total spend, requiring stricter approval controls.

- **High-Value Transactions**

```
SELECT *
FROM spend_data
```

```
WHERE total_cost > 5000;
```

- **Transaction Volume & Average Value**

```
SELECT
  COUNT(DISTINCT transaction_id) AS total_transactions,
  ROUND((AVG(total_cost))::numeric, 2) AS avg_transaction_value
FROM spend_data;
```

	total_transactions bigint	avg_transaction_value numeric
1	500	2481.16

5.4 Time-Based Spend Trends

Business Question:
How does procurement spend change over time?

Analysis Performed:
• Aggregated spend at the monthly level

Insight:
Spending patterns show periodic spikes, possibly driven by budget cycles or bulk procurement behavior.

- **Monthly Spend Trend**

```
SELECT
  DATE_TRUNC('month', purchase_date) AS month,
  SUM(total_cost) AS monthly_spend
FROM spend_data
GROUP BY month
```

ORDER BY month;

	month timestamp without time zone	monthly_spend double precision
1	2024-01-01 00:00:00	101523.94999999997
2	2024-02-01 00:00:00	78544.71
3	2024-03-01 00:00:00	123639.02999999997
4	2024-04-01 00:00:00	119348.9
5	2024-05-01 00:00:00	117917.38
6	2024-06-01 00:00:00	87118.54000000001
7	2024-07-01 00:00:00	76055.66

Total rows: 12 Query complete 00:00:00.139

5.5 Item & Category Dominance

Business Question:
Which items and categories dominate procurement spend?

Analysis Performed:
• Ranked top items by spend
• Assigned spend ranks to categories

Insight:
A limited number of items and categories drive a disproportionate share of total expenditure.

- **Top Items by Spend**

SELECT item_name,

  SUM(total_cost) AS item_spend

FROM spend_data

GROUP BY item_name

ORDER BY item_spend DESC

LIMIT 10;

	item_name text	item_spend double precision
1	Laptop	472284.81000000006
2	Annual Software Licen...	336017.9999999999
3	Monitor	130072.59000000001
4	Printer	95447.82999999997
5	Desk Chair	91857.82
6	Whiteboard	80960.79
7	Laptop Bag	20857.24
8	Stapler	6328.270000000004
9	Printer Ink	5753.889999999999
10	Notepad	999.2900000000001

Total rows: 10 Query complete 00:00:00.100

- **Category Ranking**

```
SELECT category,
  SUM(total_cost) AS category_spend,
  RANK() OVER (ORDER BY SUM(total_cost) DESC) AS spend_rank
FROM spend_data
GROUP BY category;
```

	category text	category_spend double precision	spend_rank bigint
1	Electronics	697805.2300000002	1
2	Software	336017.9999999999	2
3	Furniture	172818.61	3
4	Accessories	20857.24	4
5	Stationery	6753.1799999999985	5
6	Office Suppli...	6328.270000000004	6

5.6 Price Consistency & Variance Analysis

Business Question:
Are prices consistent across purchases?

Analysis Performed:
• Measured min–max price variance per item

Insight:
Significant price variations exist for the same items, indicating lack of pricing standardization and negotiation gaps.

- **Item Price Variance**

```
SELECT  item_name,
MIN(unit_price) AS min_price,
MAX(unit_price) AS max_price,
ROUND((
  (MAX(unit_price) - MIN(unit_price)) / MIN(unit_price) * 100)::NUMERIC,2
) AS price_variance_pct
FROM spend_data
GROUP BY item_name
HAVING MAX(unit_price) > MIN(unit_price);
```

	item_name text	min_price double precision	max_price double precision	price_variance_pct numeric
1	Desk Chair	100.9	199.59	97.81
2	Laptop Bag	20.3	46.27	127.93
3	Printer Ink	10.21	19.92	95.10
4	Annual Software Licen...	5047.01	9909.24	96.34
5	Monitor	201.29	381.95	89.75
6	Printer	150.83	292.98	94.25
7	Whiteboard	100.82	199.35	97.73
8	Laptop	806.52	994.88	23.35
9	Notepad	1.06	2.98	181.13
10	Stapler	5.35	14.43	169.72

5.7 Supplier Efficiency Analysis

Business Question:
Which suppliers are inefficient, not just high-spend?

Analysis Performed:
• Calculated spend per unit
• Benchmarked supplier prices against market averages

Insight:
Some suppliers charge above-average prices despite lower volumes, indicating inefficiencies.

- **Spend per Unit (Supplier Efficiency)**

```
SELECT supplier,
  SUM(total_cost) AS total_spend,
  SUM(quantity) AS total_quantity,
  ROUND(SUM(total_cost) / NULLIF(SUM(quantity), 0), 2) AS spend_per_unit
FROM spend_data
GROUP BY supplier
ORDER BY spend_per_unit DESC;
```

	supplier text	total_spend double precision	total_quantity numeric	spend_per_unit numeric
1	TechMart Inc.	328761.7300000001	1023	321.37
2	QuickDeliver Ltd.	285353.58999999997	1060	269.20
3	CloudSoft Corp.	200073.64000000007	806	248.23
4	OfficeSupplies ...	223580.85	931	240.15
5	FurniWorks Ltd.	202810.72000000003	1000	202.81

- **Supplier Price Benchmarking**

```
WITH item_avg_price AS (
  SELECT item_name,
    AVG(unit_price) AS market_avg_price
  FROM spend_data
  GROUP BY item_name
)
SELECT
  s.supplier,
```

```
  s.item_name,
  AVG(s.unit_price) AS supplier_price,
  i.market_avg_price,
  ROUND((
    ((AVG(s.unit_price) - i.market_avg_price) / i.market_avg_price) * 100)::NUMERIC,
    2
  ) AS inefficiency_pct
FROM spend_data s
JOIN item_avg_price i
  ON s.item_name = i.item_name
GROUP BY s.supplier, s.item_name, i.market_avg_price
HAVING AVG(s.unit_price) > i.market_avg_price
ORDER BY inefficiency_pct DESC;
```

	supplier text	item_name text	supplier_price double precision	market_avg_price double precision	inefficiency_pct numeric
1	FurniWorks Ltd.	Notepad	2.4625	1.9810416666666668	24.30
2	OfficeSupplies ...	Monitor	348.575	283.73644444444443	22.85
3	QuickDeliver Ltd.	Annual Software Licen...	8610.51375	7304.73913043478	17.88
4	QuickDeliver Ltd.	Desk Chair	163.46615384615384	146.9877358490566	11.21
5	TechMart Inc.	Laptop Bag	35.16583333333333	32.13089285714286	9.45
6	FurniWorks Ltd.	Desk Chair	158.31	146.9877358490566	7.70
7	QuickDeliver Ltd.	Printer Ink	16.406666666666666	15.254285714285713	7.55

Total rows: 23   Query complete 00:00:00.111

5.8 Potential Cost Savings Estimation

Business Question:
How much money can potentially be saved?

Analysis Performed:
• Compared paid prices with best available prices
• Estimated potential savings per item

Insight:
Significant cost-saving opportunities exist through supplier renegotiation and price standardization.

- **Potential Savings Estimation**

```
WITH best_price AS (
  SELECT
    item_name,
    MIN(unit_price) AS best_price
  FROM spend_data
  GROUP BY item_name
)
SELECT
  s.item_name,
  SUM((s.unit_price - b.best_price) * s.quantity) AS potential_savings
FROM spend_data s
JOIN best_price b
  ON s.item_name = b.item_name
WHERE s.unit_price > b.best_price
GROUP BY s.item_name
ORDER BY potential_savings DESC;
```

	item_name text	potential_savings double precision
1	Annual Software Licen...	103855.54
2	Laptop	43216.17000000003
3	Monitor	37680.47999999999
4	Printer	29082.629999999994
5	Desk Chair	28997.119999999995
6	Whiteboard	26215.529999999995
7	Laptop Bag	7621.639999999999
8	Stapler	3000.5699999999997
9	Printer Ink	1771.9899999999996
10	Notepad	455.5099999999998

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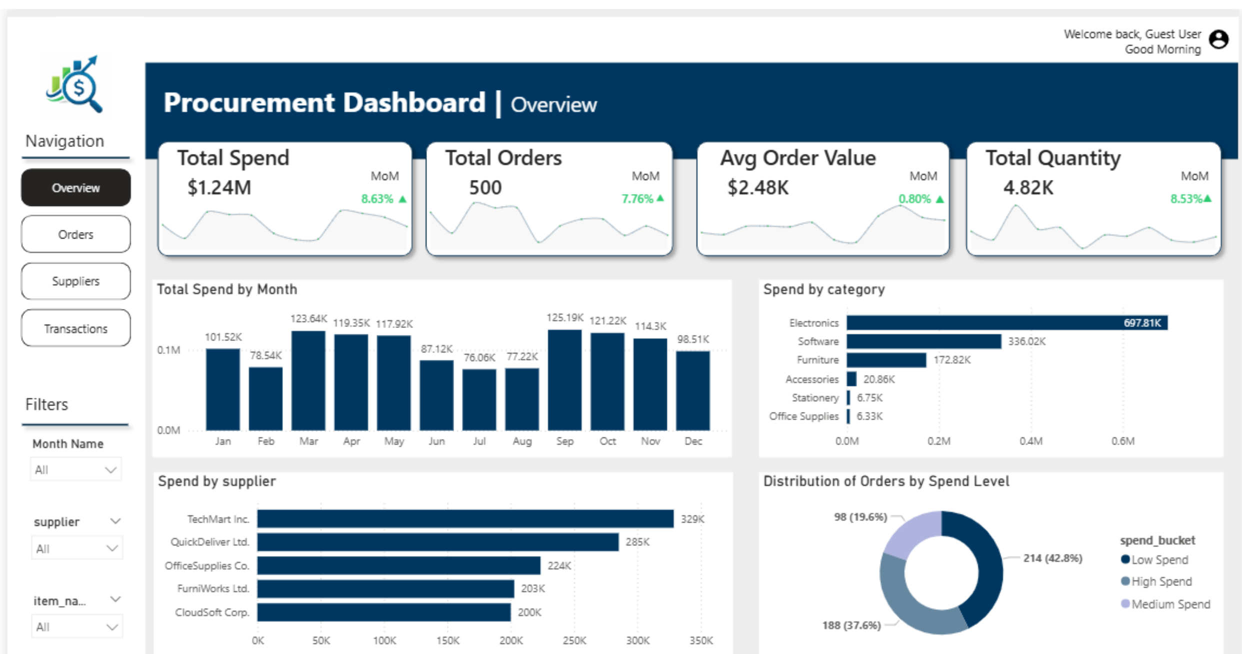
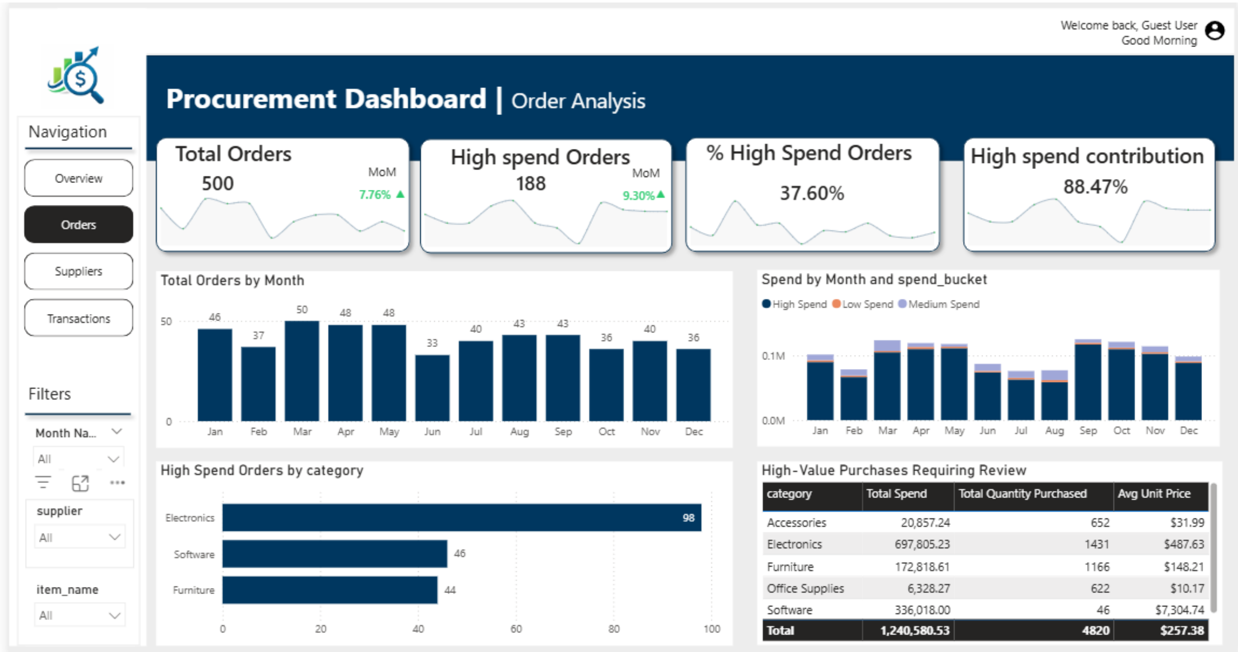
6. Power BI Dashboard & Reporting

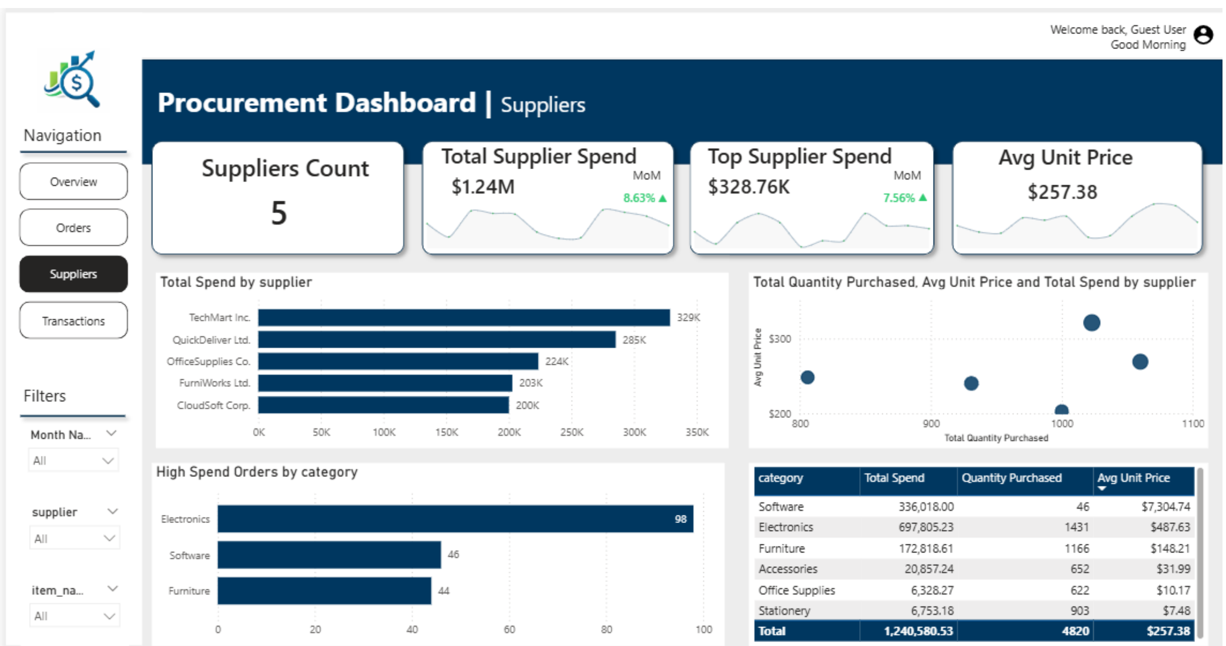
6.1 Power BI Integration

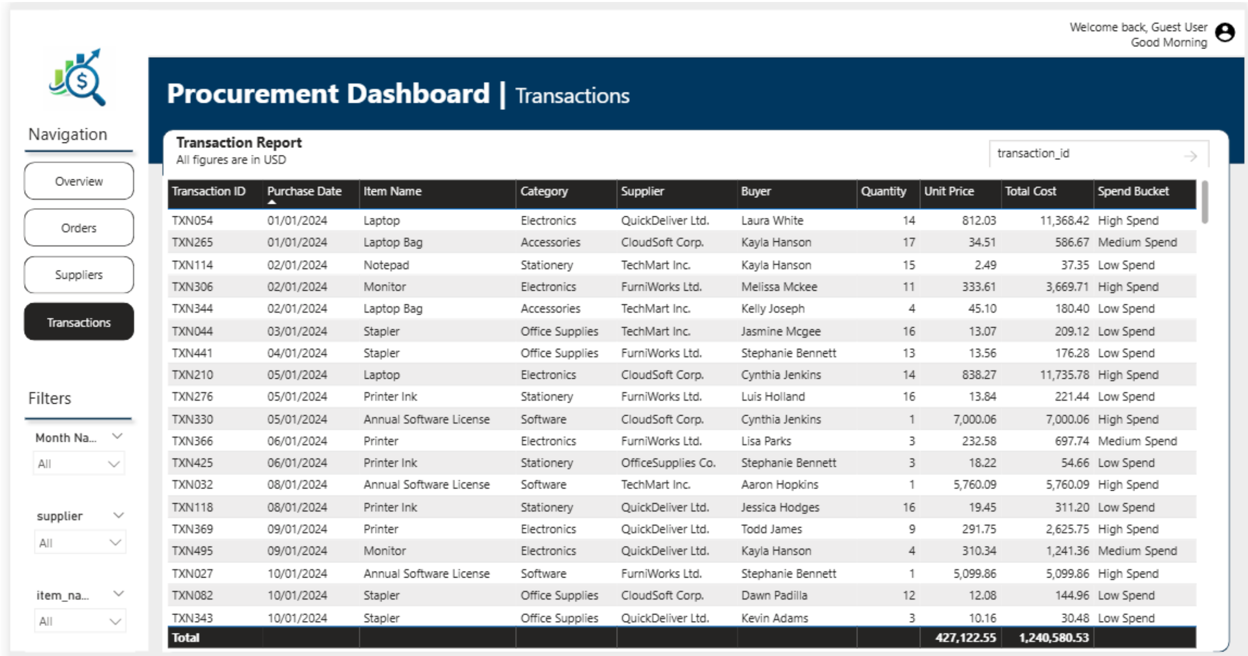
• Connected Power BI to PostgreSQL database

6.2 Dashboard Visuals




Welcome back, Guest User
Good Morning
Procurement Dashboard | Order Analysis
Navigation
Overview
Orders
Suppliers
Transactions
Filters
Month Na...
All
supplier
All
item_name
All
Total Orders
500
MoM
7.76%
High spend Orders
188
MoM
9.30%
% High Spend Orders
37.60%
High spend contribution
88.47%
Total Orders by Month
Jan 46, Feb 37, Mar 50, Apr 48, May 48, Jun 33, Jul 40, Aug 43, Sep 43, Oct 36, Nov 40, Dec 36
Spend by Month and spend_bucket
High Spend, Low Spend, Medium Spend
0.1M
0.0M
High Spend Orders by category
Electronics 98
Software 46
Furniture 44
High-Value Purchases Requiring Review
category, Total Spend, Total Quantity Purchased, Avg Unit Price
Accessories, 20,857.24, 652, $31.99
Electronics, 697,805.23, 1431, $487.63
Furniture, 172,818.61, 1166, $148.21
Office Supplies, 6,328.27, 622, $10.17
Software, 336,018.00, 46, $7,304.74
Total, 1,240,580.53, 4820, $257.38


Welcome back, Guest User
Good Morning
Procurement Dashboard | Suppliers
Navigation
Overview
Orders
Suppliers
Transactions
Filters
Month Na...
All
supplier
All
item_na...
All
Suppliers Count
5
Total Supplier Spend
$1.24M
MoM
8.63%
Top Supplier Spend
$328.76K
MoM
7.56%
Avg Unit Price
$257.38
Total Spend by supplier
TechMart Inc. 329K
QuickDeliver Ltd. 285K
OfficeSupplies Co. 224K
FurniWorks Ltd. 203K
CloudSoft Corp. 200K
Total Quantity Purchased, Avg Unit Price and Total Spend by supplier
Avg Unit Price
Total Quantity Purchased
High Spend Orders by category
Electronics 98
Software 46
Furniture 44
category, Total Spend, Quantity Purchased, Avg Unit Price
Software, 336,018.00, 46, $7,304.74
Electronics, 697,805.23, 1431, $487.63
Furniture, 172,818.61, 1166, $148.21
Accessories, 20,857.24, 652, $31.99
Office Supplies, 6,328.27, 622, $10.17
Stationery, 6,753.18, 903, $7.48
Total, 1,240,580.53, 4820, $257.38



Transaction ID	Purchase Date	Item Name	Category	Supplier	Buyer	Quantity	Unit Price	Total Cost	Spend Bucket
TXN054	01/01/2024	Laptop	Electronics	QuickDeliver Ltd.	Laura White	14	812.03	11,368.42	High Spend
TXN265	01/01/2024	Laptop Bag	Accessories	CloudSoft Corp.	Kayla Hanson	17	34.51	586.67	Medium Spend
TXN114	02/01/2024	Notepad	Stationery	TechMart Inc.	Kayla Hanson	15	2.49	37.35	Low Spend
TXN306	02/01/2024	Monitor	Electronics	FurniWorks Ltd.	Melissa Mckee	11	333.61	3,669.71	High Spend
TXN344	02/01/2024	Laptop Bag	Accessories	TechMart Inc.	Kelly Joseph	4	45.10	180.40	Low Spend
TXN044	03/01/2024	Stapler	Office Supplies	TechMart Inc.	Jasmine Mcgee	16	13.07	209.12	Low Spend
TXN441	04/01/2024	Stapler	Office Supplies	FurniWorks Ltd.	Stephanie Bennett	13	13.56	176.28	Low Spend
TXN210	05/01/2024	Laptop	Electronics	CloudSoft Corp.	Cynthia Jenkins	14	838.27	11,735.78	High Spend
TXN276	05/01/2024	Printer Ink	Stationery	FurniWorks Ltd.	Luis Holland	16	13.84	221.44	Low Spend
TXN330	05/01/2024	Annual Software License	Software	CloudSoft Corp.	Cynthia Jenkins	1	7,000.06	7,000.06	High Spend
TXN366	06/01/2024	Printer	Electronics	FurniWorks Ltd.	Lisa Parks	3	232.58	697.74	Medium Spend
TXN425	06/01/2024	Printer Ink	Stationery	OfficeSupplies Co.	Stephanie Bennett	3	18.22	54.66	Low Spend
TXN032	08/01/2024	Annual Software License	Software	TechMart Inc.	Aaron Hopkins	1	5,760.09	5,760.09	High Spend
TXN118	08/01/2024	Printer Ink	Stationery	QuickDeliver Ltd.	Jessica Hodges	16	19.45	311.20	Low Spend
TXN369	09/01/2024	Printer	Electronics	QuickDeliver Ltd.	Todd James	9	291.75	2,625.75	High Spend
TXN495	09/01/2024	Monitor	Electronics	QuickDeliver Ltd.	Kayla Hanson	4	310.34	1,241.36	Medium Spend
TXN027	10/01/2024	Annual Software License	Software	FurniWorks Ltd.	Stephanie Bennett	1	5,099.86	5,099.86	High Spend
TXN082	10/01/2024	Stapler	Office Supplies	CloudSoft Corp.	Dawn Padilla	12	12.08	144.96	Low Spend
TXN343	10/01/2024	Stapler	Office Supplies	QuickDeliver Ltd.	Kevin Adams	3	10.16	30.48	Low Spend
Total							427,122.55	1,240,580.53	

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7. Key Business Insights

- A small number of high-value transactions contribute disproportionately to total spend
- Supplier dependency presents operational risk
- Price inconsistencies highlight negotiation opportunities
- High-value transactions require governance controls
- Cost optimization potential is clearly identifiable

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8. Recommendations

- Introduce approval thresholds for high-value purchases
- Consolidate suppliers for strategic categories
- Standardize pricing benchmarks
- Monitor supplier efficiency regularly

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9. Conclusion

This project demonstrates a complete procurement analytics pipeline, transforming raw transactional data into actionable insights using Python, SQL, and Power BI. The analysis supports strategic decision-making by identifying spend drivers, risks, and optimization opportunities.

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10. Future Scope

• Predictive cost forecasting
• Supplier performance scoring
• Real-time BI integration
• Automated anomaly detection